

The AI Generalist

Prompt engineering, context engineering, projects, and settings literacy in ChatGPT and Claude

Know enough to work smarter, safer, and more deliberately



What this lecture is really about

The skill is not “using a chatbot.” The skill is shaping the work so the model can help responsibly.

Not this

A random question tossed into an empty chat, followed by blind trust in the answer.

Do this

A clear task with relevant context, output format, constraints, examples, and a check.

Result

A reusable workflow: project space, saved prompt, verified output, and a human decision.

The AI Generalist mindset

A generalist prepares context, selects a tool, checks the output, and saves what works.

Prepare

Define the job, audience, sources, and risks.

Prompt

Give role, task, format, constraints, and examples.

Compare

Use ChatGPT and Claude against the same criteria.

Verify

Check claims, assumptions, and privacy before using.

Why only ChatGPT and Claude today?

Depth beats tool collecting. Students need two familiar workbenches before chasing every new AI startup.

ChatGPT

Strong everyday workbench for drafting, project organization, files, memory, search, data tasks, Canvas-style work, and broad workflows.

Claude

Strong for long-form synthesis, careful writing, large context, Projects, Artifacts, source-aware critique, and structured thinking.

Generalist rule: choose based on the job, not brand loyalty.

Mental model: what happens when you press Enter?

Students do not need the math, but they do need the intuition.

1

Tokens: your text becomes chunks

2

Embeddings: chunks become meaning-coordinates

3

Attention: context gets weighed

4

Prediction: next tokens are generated

5

Variation: answers can differ

Prompt engineering vs. context engineering

The prompt is the order. Context is the kitchen.

Prompt engineering

The immediate instruction: role, task, format, constraints, examples, and quality standard. It improves this interaction.

Context engineering

The supporting environment: audience, files, source excerpts, previous decisions, examples, tone, data, and guardrails. It improves the whole workflow.

Prompt = what to do. Context = what the model needs to know to do it well.

The weak prompt problem

Vague prompts create vague outputs, then students blame the model.

Weak

“Help me write about AI.”

Missing: audience, purpose, source, length, tone, format, and quality criteria.

Stronger

“Write a 250-word beginner explanation of prompt engineering for first-year business students. Use a restaurant analogy, one example, and a short caution about privacy.”

The model is not psychic. It is powerful pattern completion guided by context.

A reusable prompt pattern

Identity · Context · Task · Format · Constraints · Check

Identity

Who should the model act as?

Context

What situation and audience matter?

Task

What exactly should happen?

Format

What should the output look like?

Constraints

What should be included or avoided?

Check

State assumptions or ask one question.

Examples beat adjectives

“Professional” and “clear” mean different things in different rooms.

Adjective-only instruction

“Make this polished and executive.”

The model must guess what polished means for your audience.

Example-based instruction

“Match the structure of this short memo, but write original content for my case. Keep the same heading style and level of detail.”

One good example can outperform ten vague style words.

System-style instructions vs. human prompts

Stable behavior belongs in the workspace; the day's task belongs in the prompt.

System / project instructions

Durable behavior: role, tone, safety boundaries, default format, citation habit, when to ask questions.

Human prompt

Immediate task: current goal, files, deadline, audience, examples, constraints, and deliverable.

The five layers of context

A simple recipe for getting beyond generic answers.

Identity

Who is the assistant in this situation?

World

What should it know about the course, audience, or organization?

Task

What output is needed right now?

Examples

What does good look like?

Constraints

What must be avoided, included, or verified?

Context pack ≠ file dump

Good context is selected, labeled, and connected to the task.

File dump

Everything uploaded because it might matter. The model must hunt for signal.

Context pack

Short purpose, audience, source excerpts, examples, constraints, and decision notes.

Best habit

Tell the model what each file or excerpt is and how it should use it.

PRD thinking for everyday AI

A tiny product-requirements document turns a fuzzy ask into a workable specification.

Problem

What pain are we solving?

User / audience

Who will use or read this?

Requirements

What must be true?

Output format

What should it return?

Acceptance test

How will we know it worked?

Projects: the AI Generalist's workbench

Stop scattering context across random chats.

ChatGPT Projects

A workspace for chats, files, memory behavior, project instructions, and recurring work. Useful for courses, research, weekly briefings, and writing pipelines.

Claude Projects

A self-contained workspace with project knowledge, instructions, and focused chats. Useful for long-form synthesis, critique, and reusable document workflows.

What belongs in a project?

A project should feel like a well-labeled course folder, not a messy attic.

- Purpose and intended audience
- Small source set: rubric, syllabus excerpt, notes, policy, dataset, or example
- Style sample: one or two outputs you want the model to imitate structurally
- Decision log: what has already been chosen
- Do-not-do list: private data, forbidden claims, tone boundaries
- Saved prompts and best outputs worth reusing

Project instructions create durable behavior

A good project instruction is the course policy for that AI workspace.

Sample project instruction

“Act as my AI Generalist coach. Use plain language, explain assumptions, separate source-backed facts from recommendations, ask at most one clarifying question, and never complete graded work on my behalf.”

Think of this as setting the assistant’s default manners before the work begins.

ChatGPT: useful surfaces for generalists

Projects, memory, data controls, Canvas, files, search, and apps/connectors.

Projects

Keep instructions, files, chats, and repeated work together.

Memory

Personalization can help, but students should know what is remembered.

Data controls

Training/privacy choices live in Settings; check before sensitive work.

Canvas/files

Useful when output becomes a draft, table, code sketch, or longer document.

Apps

Connect only when permission is clear.

Claude: useful surfaces for generalists

Projects, artifacts, careful synthesis, long context, and settings/capabilities.

Projects

Focused workspace with knowledge, instructions, and chat history for a durable workflow.

Artifacts

A side-panel output: app, checklist, dashboard, document, diagram, or reusable interactive object.

Writing + critique

Strong for long-form drafts, rubric-based feedback, structured memos, and synthesis across sources.

Settings are part of AI literacy

A generalist knows where personalization, privacy, and capabilities live.

Personalization

Memory, profile details, preferences, and custom instructions shape future answers.

Privacy and data

Training toggles, temporary/incognito chats, retention, exports, and deletion matter.

Capabilities

Artifacts, connectors/apps, web/search, file handling, and feature previews vary by account.

Settings map: ChatGPT vs. Claude

Labels change, but the categories stay remarkably stable.

Area	ChatGPT	Claude
Profile / personalization	Custom instructions, memory, preferred style, saved details	Profile, preferences, memory/personalization when available
Data / privacy	Data controls, temporary chat, export/delete, training toggle	Privacy controls, data export/delete, training/feedback controls
Workspace	Projects, project instructions, files, tools	Projects, project knowledge, project instructions
Creation surfaces	Canvas, files, images/tools depending on plan	Artifacts, files, capabilities depending on plan
Connections	Apps/connectors with explicit approval	Connectors/integrations depending on plan

Privacy is not paranoia; it is professionalism

Before uploading or connecting, ask what data is actually needed.

Do not upload

Private student records, health/financial data, client files, API keys, confidential employer docs.

Use instead

Fake data, excerpts, summaries, anonymized examples, public sources, or instructor-provided samples.

When unsure

Pause, ask, and use the least sensitive data that still teaches the workflow.

Source-grounded work

When accuracy matters, make the model separate source, inference, and uncertainty.

Source-backed

Directly supported by uploaded or pasted material.

Inference

Reasonable conclusion based on sources, but not directly stated.

Needs verification

Potentially current, high-stakes, ambiguous, or outside supplied sources.

The model expands the work; you own the judgment

A polished answer is not automatically a correct answer.

- Ask the model to audit its own answer for missing context and unsupported claims.
- Spot-check two or three specific claims before trusting a larger report.
- Use ChatGPT and Claude as a small “model council” for important drafts.
- Keep the human in the loop for grades, money, health, legal, and public communication.

Output becomes a workspace

The best LLM work does not end as a chat bubble.

Document

Study guide, memo, rubric, proposal, policy draft.

Interactive asset

Checklist, quiz, calculator, dashboard, prompt coach.

Reusable workflow

Project + context pack + saved prompt + review checklist.

Save your wins

Prompts are not disposable; they are workflow assets.

Save the prompt

Not just the answer. Label it by use case and model.

Save the context pattern

Which files, examples, and constraints made it work?

Save the evaluation

What did you check? What failed? What would you revise next time?

Common beginner mistakes

Most failures are workflow failures, not intelligence failures.

- Asking for too much in one prompt
- Skipping audience, format, examples, or constraints
- Uploading everything instead of selected context
- Ignoring Settings and privacy controls
- Accepting the first answer without critique
- Comparing models by vibe instead of evidence
- Letting a chat stay disposable instead of saving the workflow

A simple rubric for AI Generalist work

This is the quality bar students should internalize.

Dimension	Developing	Strong
Prompt clarity	Vague task, no format	Clear role, task, format, constraints
Context quality	Missing or file dump	Selected context pack with examples
Tool literacy	Uses chat only	Uses projects/settings/workspaces intentionally
Verification	Trusts output blindly	Checks claims, assumptions, and risks
Reusability	One-off answer	Saved workflow asset

Start free; upgrade only when the workflow proves value

A paid plan is not a substitute for prompt and context discipline.

Stay free when

You are learning, testing, drafting, and still figuring out repeatable workflows.

Consider paid when

Limits block recurring work, larger files, stronger reasoning, research, or collaboration.

Upgrade test

Will this save at least 2–3 hours this month or help ship an asset you otherwise would not ship?

Before you ask the model

The seven-question checkpoint.

- Do I know the audience?
- Did I specify the output format?
- Did I provide relevant context and examples?
- Did I state constraints and non-negotiables?
- Did I ask the model to state assumptions or uncertainty?
- Will I compare or verify the result?
- Did I save the prompt if it worked?

Recap: prompt better, context better, work better

A generalist is an operator who ships responsibly.

Prompt better

Clear instructions beat clever phrasing.

Context better

Useful context beats giant file dumps.

Work better

Projects, settings, comparison, verification, and saved workflows turn AI into a durable skill.